Regularization

Fraida Fund

Contents

Regularization . . . 2
Penalty for model complexity . . . 2
Regularization vs. standard LS . . . 2
Common regularizers . . . 2
Graphical representation . . . 3
Common features: Ridge and LASSO . . . 3
Differences: Ridge and LASSO (1) . . . 3
Differences: LASSO (2) . . . 3
Standardization (1) . . . 4
Standardization (2) . . . 4
Standardization (3) . . . 4
L1 and L2 norm with standardization (1) . . . 4
L1 and L2 norm with standardization (2) . . . 4
Ridge regularization . . . 5
Ridge term and derivative . . . 5
Ridge closed-form solution . . . 5
LASSO term and derivative . . . 5
Effect of regularization level . . . 6
Selecting regularization level . . . 6

Math prerequisites for this lecture: You should know about:

- derivatives and optimization (Appendix C in Boyd and Vandenberghe)
- norm of a vector (Section I, Chapter 3 in Boyd and Vandenberghe)

Regularization

Penalty for model complexity

With no bounds on complexity of model, we can always get a model with zero training error on finite training set - overfitting.

Basic idea: apply penalty in loss function to discourage more complex models

Regularization vs. standard LS

Least squares estimate:

$$\hat{w} = \underset{w}{\text{argmin}}\, MSE(w), \quad MSE(w) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Regularized estimate w/ **regularizing function** $\phi(w)$:

$$\hat{w} = \underset{w}{\text{argmin}}\, J(w), \quad J(w) = MSE(w) + \phi(w)$$

Common regularizers

Ridge regression (L2):

$$\phi(w) = \alpha \sum_{j=1}^{d} |w_j|^2$$

LASSO regression (L1):

$$\phi(w) = \alpha \sum_{j=1}^{d} |w_j|$$

Graphical representation

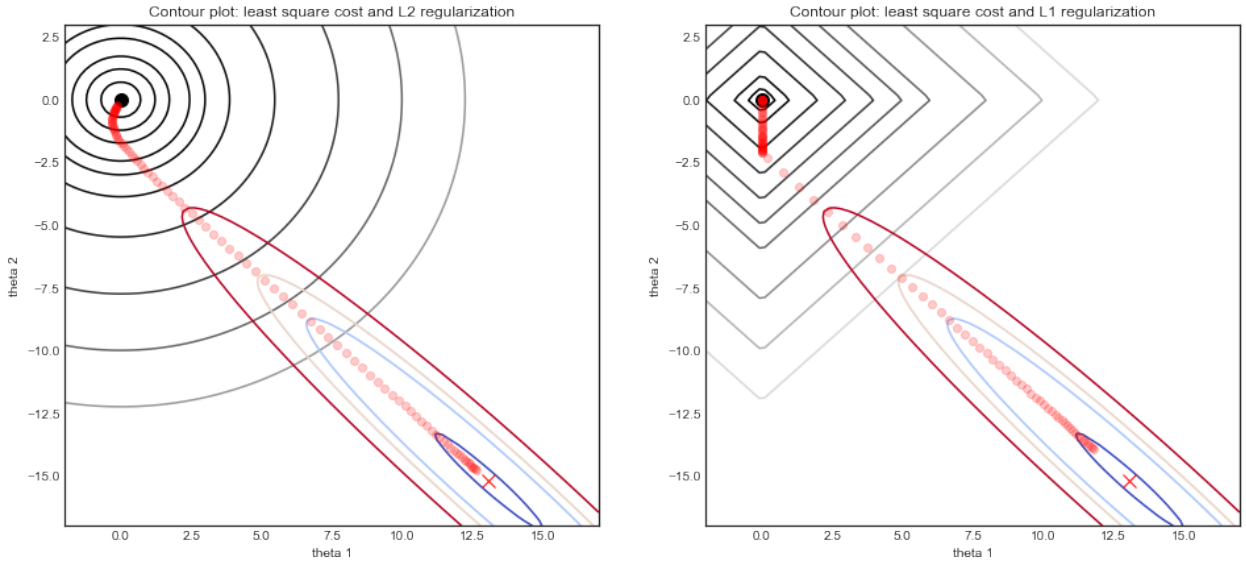


Figure 1: LS solution (+), RSS contours. As we increase α, regularized solution moves from LS to 0.

Common features: Ridge and LASSO

- Both penalize large w_j
- Both have hyperparameter α that controls level of regularization
- Intercept w_0 not included in regularization sum (starts at 1!), this depends on mean of y and should not be constrained.

Differences: Ridge and LASSO (1)

Ridge (L2):

- minimizes $|w_j|^2$,
- minimal penalty for small non-zero coefficients
- heavily penalizes large coefficients
- tends to make many "small" coefficients
- Not for feature selection

Differences: LASSO (2)

LASSO (L1)

- minimizes $|w_j|$
- tends to make coefficients either 0 or large (sparse!)
- does feature selection (setting w_j to zero is equivalent to un-selecting feature)

To understand why L1 regularization tends to make sparse coefficients but not L2 regularization - look at the graphical representation. Note that the contours of the L1 regularization "stick out" when one or both parameters is zero.

Standardization (1)

Before learning a model with regularization, we typically *standardize* each feature and target to have zero mean, unit variance:

- $x_{i,j} \rightarrow \frac{x_{i,j} - \bar{x}_j}{s_{x_j}}$
- $y_i \rightarrow \frac{y_i - \bar{y}}{s_y}$

Standardization (2)

Why?

- Without scaling, regularization depends on data range
- With mean removal, no longer need w_0, so regularization term is just L1 or L2 norm of coefficient vector

Standardization (3)

Important note:

- Use mean, variance of *training data* to transform training data
- **Also** use mean, variance of *training data* to transform **test data**

L1 and L2 norm with standardization (1)

Assuming data standardized to zero mean, unit variance, the Ridge cost function is:

$$\begin{aligned} J(\mathbf{w}) &= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^{d} |w_j|^2 \\ &= ||\mathbf{A}\mathbf{w} - \mathbf{y}||^2 + \alpha ||\mathbf{w}||^2 \end{aligned}$$

L1 and L2 norm with standardization (2)

LASSO cost function ($||\mathbf{w}||_1$ is L1 norm):

$$\begin{aligned} J(\mathbf{w}) &= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^{d} |w_j| \\ &= ||\mathbf{A}\mathbf{w} - \mathbf{y}||^2 + \alpha ||\mathbf{w}||_1 \end{aligned}$$

Ridge regularization

Why minimize $||\mathbf{w}||^2$?

Without regularization:

- large coefficients lead to high variance
- large positive and negative coefficients cancel each other for correlated features (remember attractiveness ratings in linear regression case study...)

Ridge term and derivative

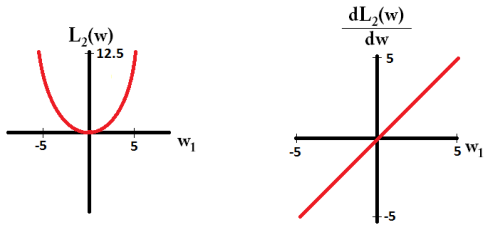


Figure 2: L2 term and its derivative for one parameter.

Ridge closed-form solution

$$J(\mathbf{w}) = ||\mathbf{A}\mathbf{w} - \mathbf{y}||^2 + \alpha||\mathbf{w}||^2$$

Taking derivative:

$$\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} = -2\mathbf{A}^T(\mathbf{y} - \mathbf{A}\mathbf{w}) + 2\alpha\mathbf{w}$$

Setting it to zero, we find

$$\mathbf{w}_{\text{ridge}} = (\mathbf{A}^T\mathbf{A} + \alpha\mathbf{I})^{-1}\mathbf{A}^T\mathbf{y}$$

LASSO term and derivative

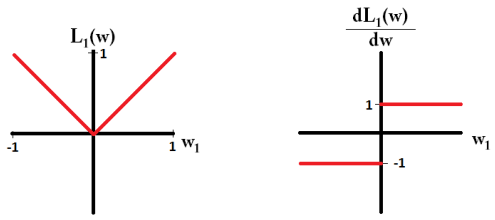


Figure 3: L1 term and its derivative for one parameter.

- No closed-form solution: derivative of $|w_j|$ is not continuous
- But there is a unique minimum, because cost function is convex, can solve iteratively

Effect of regularization level

Greater α, less complex model.

- Ridge: Greater α makes coefficients smaller.
- LASSO: Greater α makes more weights zero.

Selecting regularization level

How to select α? by CV!

- Outer loop: loop over CV folds
- Inner loop: loop over α